

REY AUDIE S. ESCOSIO

Ph.D. student studying infectious disease modelling

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📍 Lisbon, Portugal

External links: [in](#) LinkedIn · [G](#) Google Scholar · [R](#)⁶ ResearchGate · [G](#)itHub · [ID](#) ORCID

Research interests:

infectious disease modelling

mathematical biology

statistical inference

optimization

complex systems

EDUCATION

Doctor of Philosophy in Physics and Astrophysics (Physics)

Faculty of Sciences, University of Lisbon

📅 Jan 2024 – Present

📍 Lisbon, Portugal

Project: *Targeted control of COVID-19 post mass vaccination*

- Focused on the research fellowship under the supervision of Dr. Ganna Rozhnova and Dr. Ana Nunes
- Researcher under the BioSystems and Integrative Sciences Institute, FCUL and visiting researcher in the Infectious Disease Modeling Group, University Medical Center Utrecht, Utrecht University

Master of Science (Applied Mathematics) (Mathematics in Life and Physical Sciences)

Institute of Mathematics, University of the Philippines Diliman

📅 Aug 2018 – Jan 2021

📍 Quezon City, Philippines

- Department of Science and Technology-Accelerated Science and Technology Human Resource Development Program - National Science Consortium (DOST-ASTHRDP) Scholarship awardee

Thesis: *A perturbed accelerated gradient descent algorithm using an n -dimensional golden section search method*

- Student researcher under the supervision of Dr. Renier Mendoza (Jan 2019 – Jan 2021).

Bachelor of Science in Applied Physics (Instrumentation Physics)

National Institute of Physics, University of the Philippines Diliman

📅 Jun 2013 – Jun 2018

📍 Quezon City, Philippines

- Department of Science and Technology-Science Education Institute (DOST-SEI) RA 7687 Scholarship awardee

Thesis: *Response of single and connected Hodgkin-Huxley neurons to a sinusoidal input*

- Student researcher in Complexity Science Group under the supervision of Dr. Johnrob Bantang (Oct 2015 – May 2018)
- Presented in three Samahang Pisika ng Pilipinas (SPP) Physics Conferences (2016 – 2018) accessible [here](#) ➡.

RESEARCH

Publications

1. Canfora, B., Escosio, R. A. S., et al., (2025). Retrospective evaluation of school-related measures on pre-vaccination transmission dynamics of SARS-CoV-2. *Communications Medicine* ➡. (Shared first co-authorship).
2. Canfora, B., Escosio, R. A. S., et al., (2026). Counterfactual evaluation of elementary and secondary school policies in the COVID-19 pandemic. *Nature Communications* ➡. (Shared first co-authorship).
3. Westerhof, I., Escosio, R. A. S., et al., (2026). Model-informed post-pandemic COVID-19 age and risk stratified vaccination strategies under hybrid immunity landscapes in Norway. (In preparation).
4. Escosio, R. A. S., & Mendoza, R. G., (2025). Accelerated gradient descent algorithm with Golden Section search for escaping saddle points in optimization problems. *Matimyas Matematika* ➡.
5. Mata, M. A. E., Escosio, R. A. S., et al., (2024). Analyzing the dynamics of COVID-19 transmission in select regions of the Philippines: a modeling approach to assess the impact of various tiers of community quarantines. *Heliyon* ➡. (Shared first co-authorship).
6. Escosio, R. A. S., et al., (2023). A model-based strategy for COVID-19 vaccine roll-out in the Philippines. *Journal of Theoretical Biology* ➡.

Selected conference presentations

1. *The elasticity of SARS-CoV-2 transmission across the demographic landscape*. XLVI Dynamics Days Europe (DDE 2026), July 19-24, 2026. (oral presentation)
2. *The elasticity of SARS-CoV-2 transmission across the demographic landscape*. 14th European Conference on Mathematical & Theoretical Biology (ECMTB 2026), July 13-17, 2026. (oral presentation)
3. *Quantifying the role of age in COVID-19 transmission*. 8th International Workshop on Mathematical Biology (IWOMB 2026), January 6-9, 2026. (poster presentation)

- **Won the student poster prize, 2nd place**
- 4. *Quantifying the role of age in COVID-19 transmission*. 10th International Conference on Infectious Disease Dynamics (EPIDEMICS 10), November 30-December 3, 2025. (poster presentation)
- 5. *Characterizing sociodemographic influences to the reproduction number of SARS-CoV-2*. 16th Dynamical Systems applied on Biology and Natural Sciences (DSABNS 2025), January 20-24, 2025. (poster presentation)
- 6. *The what-ifs of early COVID-19 transmission in the constituent units of the National Capital Region, Philippines*. Society of Mathematical Biology 2022 Annual Meeting (SMB 2022), September 19-23, 2022. (online poster presentation)
- 7. *Modelling COVID-19 dynamics with different community quarantine protocols in National Capital Region, Philippines*. Society of Mathematical Biology 2021 Annual Meeting (eSMB 2021), virtual meeting, June 13-17, 2021. (poster presentation)
- 8. *Parameter estimation of the Fitzhugh-Nagumo model via a perturbed accelerated gradient descent algorithm with an n -dimensional Golden Section search method*. Society of Mathematical Biology 2020 Annual Meeting (eSMB 2020), virtual meeting, August 17-20, 2020. (poster presentation)
 - **Won the subgroup student poster prize**

Speaking engagements

1. *Retrospective evaluation of school-related measures on pre-vaccination transmission dynamics of SARS-CoV-2*. Young Complexity Researchers' Lunch (YCRL), Centre for Complex Systems Studies at Utrecht University, September 1, 2026.
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Selected participation

1. International Workshop on Mathematical Biology (IWOMB). (January 22-25, 2024; July 26-27, 2023; June 9-11, 2021; January 6-8, 2020; January 6-10, 2019)
2. **Unifying the Epidemiological and Evolutionary Dynamics of Pathogens** Program. Nordic Institute for Theoretical Physics, Stockholm, Sweden. May 29-June 23, 2023 (attended first two weeks).
3. **Spring College on the Physics of Complex Systems**. International Center for Theoretical Physics, Trieste, Italy. February 20-March 17, 2023
4. SEAMS School 2022 on Modern Trends in Signal and Data Processing. Quezon City, Philippines. December 5-13, 2022.
5. ICTP Asian Network School and Workshop on Condensed Matter and Complex Systems 2019. Quezon City, Philippines. November 4-8, 2019.
6. 16th International Conference on Molecular Systems Biology (ICMSB 2019). Manila, Philippines. October 28-31, 2019.

EXPERIENCE

Research assistantship

 Nov 2020 – Dec 2022

 Quezon City, Philippines

- **University Research Associate**. Natural Sciences Research Institute, University of the Philippines Diliman (Jan 2022 – Dec 2022).
 - **Science Research Specialist and Consultant**. Center for Applied Modeling, Data Analytics, and Bioinformatics for Decision Support Systems in Health (Nov 2021 – Dec 2022).
 - **Research Associate**. Modelling and Applications Group, Institute of Mathematics, University of the Philippines Diliman (May 2021 – Nov 2021).
 - **Project Assistant**. Resilience Institute, University of the Philippines (Nov 2020 – May 2021).
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Teaching and supervision

 Sep 2021 – Jul 2026

 Lisbon, Portugal and Quezon City, Philippines

- **Part-time teacher** of undergraduate experimental physics class. Department of Physics, Faculty of Sciences, University of Lisbon (Feb 2026 – Jul 2026).
- **Co-supervisor** of B.Sc. Mathematics student Christine Ann Tongol. Institute of Mathematics, University of the Philippines Diliman (Sep 2021 – May 2022).
- **Lecturer** of undergraduate precalculus classes. Institute of Mathematics, University of the Philippines Diliman (Sep 2021 – Jan 2022).

SKILLS

-  **Programming and scientific computing**
Proficient in Python, MATLAB, Wolfram Mathematica, and R; working knowledge of Stan, C++, and SQL; familiar with HPC environments and the GitHub environment
-  **Typesetting and other software**
Effective in using $\text{\LaTeX}$; proficient in MS Office and similar software
-  **Languages**
English and Filipino – proficient in both oral and written communication